

$$f(x) = \begin{cases} \sin 2x & x > 0 \\ a + bx & x \leq 0 \end{cases}$$

For continuity,

$$f(0) = f(0^+)$$

$$\Rightarrow a = 0 \quad - (1)$$

For differentiability,

$$f'(0^-) = f'(0^+)$$

$$\Rightarrow \text{[scribbled out]}$$

$$\Rightarrow 2 \cos(0) = b$$

$$b = 2 \quad - (2)$$

$$\Rightarrow a + b = 2$$